Channel-Set Drift in Long-Reasoning W4A16 LLMs: Mechanisms, Failed Remedies, and Budget-Dependent Signals

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Abstract

Static per-channel protection is attractive for W4A16 reasoning inference, but it assumes that high-magnitude activation channels remain useful protection targets throughout decode. We measure decode-time channel-set drift across four reasoning LLMs and find strict top-1% set-leaving of 53–67% by horizons up to 20K tokens, with a MATH-500 replication within 0.02 of the AIME-2025 rates on three measured models. We observe the effect at the measured block-output surface in hybrid Mamba-2, parallel-hybrid, and pure-Transformer models, extending DecDEC's short-horizon Transformer observation and showing that Quamba2's channel-order and activation-persistence observations do not imply stable top-K block-output protection at this measured surface. We then test eight channel-set protection families, a rotation baseline, and a rotation-plus-budget composition with matched-cost and random controls. Hard position or scale switches are actively harmful, top-1% smoothing and cross-tensor coupling have signal but insufficient recovery, and reactive DecDEC-style selection does not recover the W4A16 gap. The strongest channel-set signal is budget-dependent: on Nemotron-3, top-10 EMA protection recovers 0.815 of the BF16-vs-static gap with a 0.220 descriptive margin over static top-10; on Granite-Small, top-5 recovers 0.449 with a wide CI. A ParoQuant-style rotation baseline recovers 0.754 on Granite-Small, while naive ParoQuant+M11b composition is sub-additive. KL trajectories grow sublinearly, with 100-token autocorrelation and high spectral entropy on DeepSeek-R1-Distill and Falcon-H1, weakening compound-error explanations in the measured regime.

1 Introduction

Long-reasoning models can spend thousands of decode steps in a single answer, so calibration-time quantization decisions must survive within-trace distribution shift. Static protected-channel maps are especially brittle if the channels that dominate early activation magnitude do not remain dominant later. This paper studies that assumption for W4A16 post-training quantization (PTQ): 4-bit weights, 16-bit activations, and a small protected activation-channel set kept at higher precision.

The assumption matters for recent state-space quantization. Quamba2 reports channel-order preservation in selective-SSM computation and channel/state persistence for SSM activations. Those observations motivate its offline sort-and-cluster weight reordering and per-state-group quantization (Chiang et al., 2025). We test the corresponding protected-set assumption at long reasoning horizons. Across Granite-4.0-H-Small, NVIDIA-Nemotron-3-Nano-30B-A3B-BF16 (Nemotron-3), DeepSeek-R1-Distill-Qwen-1.5B, and Falcon-H1, 53–67% of the top-1% channels at the calibration position strictly leave the later top-1% set. The channels do not merely shuffle rank within the set; they leave the set a static policy would protect.

We target a narrow but practical systems problem. Reasoning workloads are expensive, and inference-cost forecasts expect large pressure to reduce per-token cost even as reasoning demand grows (Wiggers, 2025; Gartner, 2026). Recent low-bit PTQ systems for larger standard Transformers can recover substantial quality, including ParoQuant's weight-only scaled-pairwise-rotation reasoning results (Liang et al., 2026). Our focus is different. We study small reasoning models, hybrid Mamba-2 or parallel-hybrid architectures, pure PTQ rather than distillation, and channel-set protection under long decode.

Our contributions are fourfold.

- Empirical drift. We measure decode-time channel-set drift across four models and architectures up to 20K tokens, finding strict set-leaving of 53–67% across hybrid Mamba-2, parallel-hybrid, and pure-Transformer models, with MATH-500 replication on three models.
- Prior-art differentiation. We extend DecDEC's short-horizon Transformer evidence to the long-reasoning regime and show that Quamba2-style protected-set persistence does not hold at the measured block-output surface for the hybrid models tested here.
- Mechanism tests. We evaluate eight channel-set protection families with random and matched-cost controls and identify three mechanisms: boundary discontinuities are harmful, smooth top-1% tracking is insufficient, and protected budget determines whether the method recovers quality; KL/FFT diagnostics on Granite, DeepSeek, and Falcon do not support runaway compound error in the measured packets.
- Budget-dependent signal and scope. Budget-tuned EMA protection succeeds on Nemotron at top-10 and recovers 0.449 on Granite top-5 with wide uncertainty, while ParoQuant recovers 0.75 on Granite via rotation. Naive composition is sub-additive, scoping the positive path toward method selection rather than simple stacking.

2 Related Work

Mamba and state-space quantization. Vision-Mamba quantization already reports dynamic activation outliers. QMamba introduces temporal grouping for hidden states (Li et al., 2025), and OuroMamba uses dynamic outlier detection during inference (Ramachandran et al., 2025). Language-Mamba work often assumes more stable structure. Mamba-PTQ identifies recurrent outlier channels (Pierro and Abreu, 2024); Quamba and Quamba2 use static or calibration-time structure for Mamba-family PTQ (Chiang et al., 2024; 2025); MambaQuant uses variance-aligned rotations (Xu et al., 2025). We ask whether the static protected-set assumption holds in long reasoning traces.[1]

LLM quantization and rotations. SmoothQuant, AWQ, QuaRot, KVQuant, and BlockDialect protect or transform activation structure using calibration-time statistics, rotations, or adaptive formats (Xiao et al., 2023; Lin et al., 2023; Ashkboos et al., 2024; Hooper et al., 2024; Jang and Tambe, 2025). CMPQ allocates per-channel precision from calibration-time activation distributions; our work tests whether calibration-time channel selection remains stable under long-decode drift (Chen et al., 2025). ResQ keeps a low-rank PCA subspace at higher precision throughout decode, while we measure whether channel-basis top-K protection is stable across decode positions and find it often is not (Saxena et al., 2025). Rotated Runtime Smooth is the closest rotation-plus-runtime-smoothing adjacency. It uses runtime channel/group-wise activation maxima as smoothing scales in fused GEMM rather than maintaining per-layer top-K EMA protection sets (Yi et al., 2024). Our results do not contradict these systems. The narrower implication is that, when protected channel sets change during long decode, static channel maps need architecture- and horizon-specific validation. ParoQuant is particularly important because it reports strong reasoning-task accuracy under 4-bit weight-only PTQ using scaled independent Givens/pairwise rotations plus channel-wise scaling (Liang et al., 2026); its project page reports Qwen3-4B AIME-24

[1] We use decode-time channel-set drift for the measured phenomenon. SmoothQuant's "migration" is activation-to-weight scale transfer (Xiao et al., 2023); MoBiQuant uses related language for token-set shifts across bit-widths (Wang et al., 2026).

scores of AWQ 62.2, ParoQuant 73.3, and FP16 75.6 (Z Lab, 2026). Our Granite ParoQuant-style baseline tests whether a rotation mechanism, rather than channel-set protection, can recover the endpoint measured here.

Dynamic and long-context policies. DecDEC dynamically identifies salient activation channels at each decode step and reports roughly 20% recall for static outlier-channel identification against per-step ground truth in short Transformer decode (Park et al., 2025). We extend that setting in four ways: longer reasoning horizons, strict set-membership drift accounting, hybrid Mamba/parallel-hybrid coverage, and reasoning-trace intervention controls. The pure-Transformer result serves as a falsification/control surface. KV-cache methods such as KIVI, PM-KVQ, AttentionPredictor, ChanMix, and MixKVQ adapt or compress cached key/value tensors (Liu et al., 2024; 2025; Yang et al., 2025; Liao and Wen, 2026; Zhang et al., 2025); our interventions target linear-layer activation channels.[2]

Error accumulation. SLQ shows that layer-wise non-uniform/asymmetric quantization with wide bitwidth search can be task-lossless below 4 bits per parameter and distribution-lossless at roughly 5–6 bits per parameter (Helcig et al., 2026). QEP emphasizes cross-layer quantization-error growth (Arai and Ichikawa, 2025). LAQuant emphasizes layer-wise lookahead losses and calibration/deployment alignment for reasoning models (Choi et al., 2026). Activation Sensitivity formalizes channel-wise perturbation impact on loss and downstream propagation (Xu, 2026). These results motivate testing whether our more aggressive W4A16 endpoint exhibits decode-position error growth rather than assuming it.

3 Method

For each trace, layer, and measured decode position, we record the vector of per-channel activation magnitudes. Let $S_{\ell,p}(t)$ be the top-1% channel set for layer ℓ and prompt p at position t. The strict set-leaving rate at position t is

$$1 - \frac{|S_{\ell,p}(100) \cap S_{\ell,p}(t)|}{|S_{\ell,p}(100)|}, \tag{1}$$

averaged over prompts and layers. Within-set shuffling is reported separately for channels that remain in the top-1% set but move substantially in rank.

We score interventions using recovery of the BF16-vs-static W4A16 gap:

$$R = \frac{L_{\text{static}} - L_{\text{method}}}{L_{\text{static}} - L_{\text{BF16}}}. \tag{2}$$

Recovery may be negative when a method is worse than static protection. It may exceed one when a method beats BF16 on the scored slice. We report unclipped medians and bootstrap CIs. We explicitly track traces with no recoverable static gap and exclude them from ratio computation for method CIs; their frequency is part of the trace-heterogeneity result.

The method classes are grouped by the mechanism they test: static coverage, hard switching, smoothing, cross-tensor signal, reactive selection, larger-budget tracking, and stable cores. We test eight channel-set method classes: static migration-aware unions, hard position-conditioned protected sets, hard position-binned scale tables, EMA-smoothed top-magnitude protection (M11), cross-tensor activation+key coupling (M18), DecDEC-style reactive top-1% selection, budget-tuned EMA protection (M11b), and stable-core protection (M26). We compare each method to static baselines plus random or matched-budget controls when applicable. We evaluate ParoQuant as an algorithmic rotation baseline, not as a channel-set method, and test one direct ParoQuant+M11b composition after both individual mechanisms showed signal.

[2] "Hot channels" in HCP/CHON refers to persistently extreme channels during NVFP4 pretraining (Dong et al., 2026). This training-step time axis complements our decode-step time axis: channels can become persistent across pretraining iterations while drifting during a single inference trajectory. We refer to Koike-Akino et al. (2026) as test-time TTQ to avoid conflict with classical Trained Ternary Quantization.

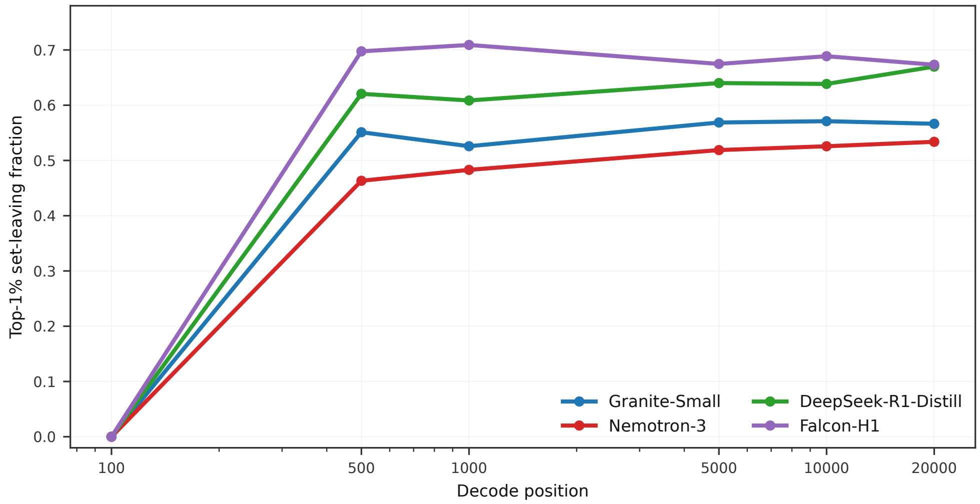

Figure 1: Strict top-1% channel-set leaving relative to decode position 100. Static channel protection loses 53–67% of its original protected set by the final measured position across four reasoning models.

4 Experiments

Models and traces. The measurement suite covers Granite-4.0-H-Small, Nemotron-3, DeepSeek-R1-Distill-Qwen-1.5B, and Falcon-H1. Intervention experiments use 12 deterministic long-reasoning traces except where stated otherwise; the original drift measurement uses 24 Granite-Small traces. We measure decode positions on the grid $\{100, 500, 1000, 5000, 10000, 20000\}$ when the model packet supports 20K tokens. The cross-prompt replication uses 30 MATH-500 prompts on Granite-Small, DeepSeek-R1-Distill, and Falcon-H1; GPQA-Diamond was gated and not used for claims. Dense KL analysis uses a Granite-Small packet with all positions 1–512, every 10th position to 2K, every 50th position to 10K, and every 100th position to 20K. The narrowed cross-model diagnostic repeats the KL/FFT summary on DeepSeek-R1-Distill and Falcon-H1.

Controls. Static top-1% protection is the baseline. Position- and scale-switching methods use random-bin controls. EMA and reactive methods use random-walk or random-reactive controls. Budget experiments include matched static top-10 controls. The controls are decisive: two methods that look plausible in isolation are worse than their random controls.

5 Results

5.1 Channel sets drift at long reasoning horizons

Figure 1 shows the headline measurement. The traces are not monotone at every intermediate point. Even so, all four models have large strict set-leaving by the final measured position: 0.566 for Granite-Small, 0.534 for Nemotron-3, 0.671 for DeepSeek-R1-Distill, and 0.674 for Falcon-H1. The pure-Transformer reference drifts at least as much as the hybrid models, so the drift extends beyond Mamba layers.

The effect is not an AIME-only artifact in the models where we reran the measurement. On 30 MATH-500 prompts, strict set-leaving is 0.561 for Granite-Small, 0.651 for DeepSeek-R1-Distill, and 0.675 for Falcon-H1. Each is within 0.02 absolute of its AIME-2025 rate, passing the preregistered cross-prompt tolerance. GPQA-Diamond was not included because the local benchmark access path was gated during this run.

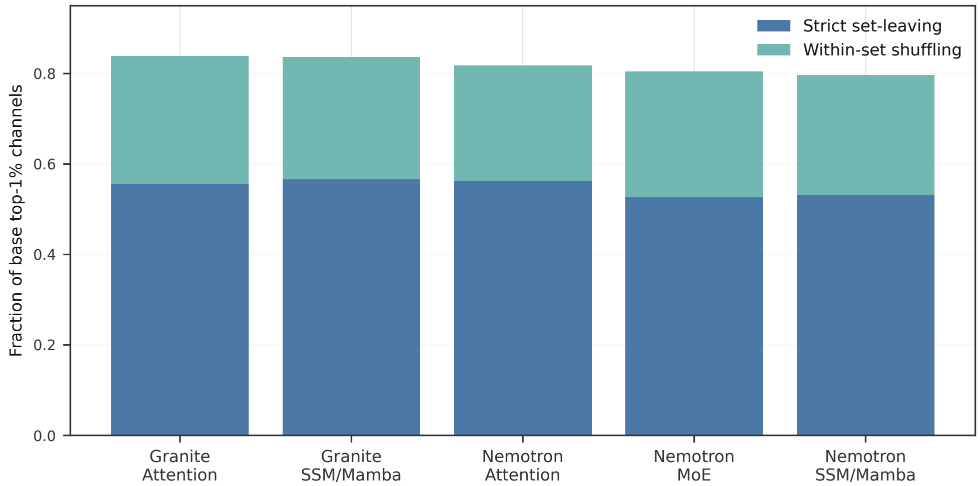

Figure 2: Layer-stratified drift at measured block outputs. Attention, SSM/Mamba, and MoE-classified layers show comparable strict set-leaving, so no single component type localizes the drift.

5.2 Layer type does not explain the drift away

Because the Quamba2 comparison turns on persistence in SSM computation, the relevant observable here is whether drift reaches SSM/Mamba block outputs rather than only attention or MLP outputs. Figure 2 shows that Granite and Nemotron block-output drift is comparable across layer types. Granite attention and SSM/Mamba outputs have strict set-leaving of 0.557 and 0.567. Nemotron attention, MoE, and SSM/Mamba outputs have 0.563, 0.527, and 0.533. This does not isolate internal SSM state tensors, but it shows that the measured block-output surface does not preserve channels.

5.3 Top-1% channel-set interventions fail despite control signal

Figure 3 and Table 1 summarize the intervention outcomes. The random-bin control beats hard position-conditioned protection by 0.668 median recovery, and random scale-bin assignment beats hard scale bins by 0.761. EMA smoothing is different: M11 beats its random-walk control by 1.25 median, but recovers only 0.048 in absolute terms. M18 activation+key coupling has the strongest signal-vs-random margin among failed top-1% methods, beating random coupling by 4.32 median. Its absolute recovery is still negative. DecDEC-style reactive selection beats a random reactive control by 1.19 median, but still has negative recovery and does not recover quality. M26 stable-core protection shows signal over random matched-core channels but remains below the positive-method threshold.

5.4 Budget helps, but rotation remains the stronger Granite baseline

M11b turns a uniformly negative channel-set picture into a model-dependent positive result, while ParoQuant sets the stronger Granite baseline. On Granite-Small, top-5 EMA protection recovers 0.449 of the static gap and beats static top-10 by 0.501 median, but the BCa CI is wide. Nemotron top-5 has a strictly positive BCa CI, yet it does not beat static top-10. The top-10 arm is the strongest channel-set result: median recovery 0.815, BCa CI [0.158, 0.906], Holm-adjusted $p = 0.030$, and a 0.220 median margin over static top-10. Budget is a real mechanism. The best budget and the value over static protection are model-dependent.

ParoQuant constrains the claim. Its Granite-Small rotation baseline recovers 0.754, exceeding M11b top-5 on the same model, though this descriptive high-median signal is not a Holm rejection within the 33-test E15 family. The paper does not argue that W4A16 PTQ is intrinsically broken; it argues that channel-set protection is fragile under long reasoning

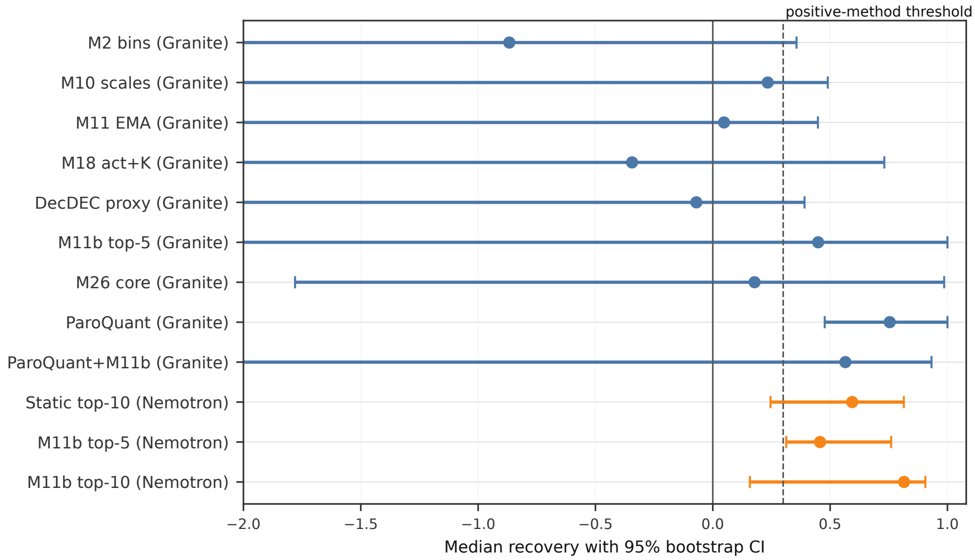

Figure 3: Median recovery with 95% bootstrap CIs. The plot clips long negative CI whiskers at the boundary for readability; Table 1 reports exact rounded values. The dashed line marks the 0.30 positive-method threshold used in preregistered gates.

Method family	Model	Median recovery	95% CI	Key comparison
Position-conditioned channels	Granite	-0.867	[-4.25, 0.357]	random bin better by 0.668
Position-binned scales	Granite	0.234	[-3.47, 0.490]	random scale bin better by 0.761
EMA-smoothed top-1%	Granite	0.048	[-8.94, 0.448]	random-walk lower by 1.25
Activation+key coupling	Granite	-0.344	[-14.1, 0.731]	random coupled lower by 4.32
DecDEC proxy	Granite	-0.070	[-8.50, 0.391]	random reactive lower by 1.19
Stable core	Granite	0.178	[-1.78, 0.986]	random matched-core lower by 1.73
Budget EMA top-5	Granite	0.449	[-2.28, 1.00]	static top-10 lower by 0.501
Static top-10	Nemotron	0.594	[0.246, 0.814]	Holm-significant matched-budget control
Budget EMA top-5	Nemotron	0.457	[0.313, 0.760]	Holm-significant, below static top-10
Budget EMA top-10	Nemotron	0.815	[0.158, 0.906]	Holm-significant; static top-10 lower by 0.220
ParoQuant rotation	Granite	0.754	[0.477, 1.00]	M11b top-5 lower by 0.304
ParoQuant+M11b top-10	Granite	0.565	[-63.6, 0.932]	trails ParoQuant by 0.189

Table 1: Intervention outcomes with BCa 95% CIs where exact per-trace recovery values are available. We leave recovery unclipped; values can be negative or exceed one. The Holm tests evaluate median recovery greater than zero across the method-table family; paired method-vs-control margins are descriptive unless separately tested. ParoQuant is included as a rotation baseline and is not a channel-set method.

drift, while rotation-based conditioning can address part of the same endpoint by a different mechanism. That distinction matters.

The direct composition test was negative. On Granite-Small, ParoQuant+M11b top-10 recovers 0.565, trailing ParoQuant alone by 0.189 median recovery and also trailing the ParoQuant+random top-10 control. This rules out the simplest "stack the two remedies" story. Rotation appears to absorb much of the recoverable channel-set signal in this setting, so future positive methods need a decision rule or a more coupled design rather than naive composition.

5.5 KL does not support runaway compounding

Figure 4 tests whether failed channel-set methods primarily fail because errors compound over decode. In the dense Granite-Small packet, static top-1%, DecDEC proxy, and M11 have lowest residuals under a sublinear square-root fit among the tested linear, sublinear, and superlinear alternatives. Their mean KL values are 0.150, 0.133, and 0.131 respectively. Their fitted AR decays are 0.51, 0.45, and 0.44.

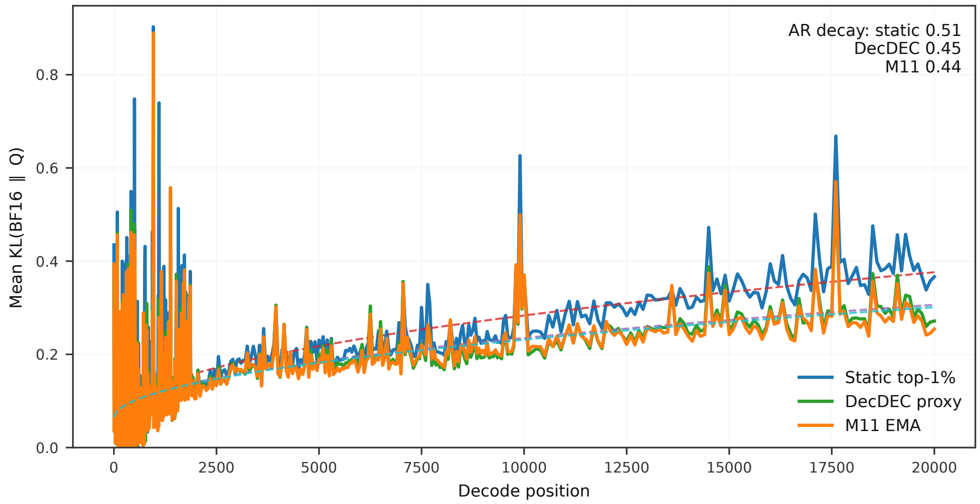

Figure 4: Per-position KL trajectories on Granite-Small. Among the tested functional forms, all three regimes have lowest residuals under sublinear square-root fits and moderate AR decay, rather than superlinear growth.

The narrowed cross-model diagnostic gives the same qualitative answer on the two smaller non-Granite models. DeepSeek-R1-Distill and Falcon-H1 both select sublinear square-root fits for static top-1%, DecDEC proxy, and M11. Their activation spectra are broadband rather than periodic: median normalized spectral entropy is 0.850 for DeepSeek and 0.891 for Falcon, with 100-token autocorrelation in both packets. This weakens compound error as the primary explanation in the measured regime and shifts attention back to stale or misbudgeted protection.

6 Discussion

The results point to three mechanisms. First, boundary discontinuities are actively harmful: both hard protected-set and hard scale-bin switching lose to random controls. Salfati's KV-cache analysis provides a compatible formal analogy: under the softmax Fisher metric, dropping a direction is quadratically worse than quantizing it at the same bit cost (Salfati, 2026). Second, smoothing removes the worst boundary harm but does not recover quality at top-1% budget; M11 and M18 contain signal, but not enough protected capacity. Third, budget is binding, but budget alone is not a universal recipe: Nemotron top-10 succeeds, Granite top-5 recovers 0.449 with wide uncertainty, and static top-10 is already strong on Nemotron.

Trace heterogeneity remains important. In most Granite static-top-1% channel-set experiments, roughly one third of the 12-trace slice has no recoverable static gap under the ratio metric; in the larger static-union experiment the no-gap fraction is 0.375. Early static-union gates retain no-gap traces as zero-recovery rows, while later method packets report recovery ratios on positive-gap traces and track the no-gap fraction separately. This limits its median-based claims and explains the wide CIs on several methods. After BCa/Holm correction over 33 method-family tests, corrected-positive channel-set evidence is limited to Nemotron budgeted EMA top-5/top-10; paired method-vs-control margins remain descriptive unless separately tested. The paper emphasizes controlled mechanism comparisons and cross-model directionality rather than a single universal method headline.

The saturated branches are hard switching, top-1% smoothing alone, DecDEC-style top-1% reactive selection, stable-core protection alone, top-1% activation+key coupling, and naive rotation-plus-budget composition. ParoQuant shows that rotations can recover much of the Granite W4A16 gap; M11b shows that protected budget matters on Nemotron. The failed composition is useful because it separates "both mechanisms have signal" from "their effects add." The next positive-method branch should choose between mechanisms from measurable model properties or couple rotation and channel selection more tightly.

7 Limitations

We scope the study to W4A16 PTQ on small reasoning models and measured block-output activation channels. We do not claim results for FP8, MXFP4/NVFP4, W4A4, or quantization-aware training; the format-axis probe found available FP8/W4A16 paths but no benchmark-metrics adapter, and no importable NVFP4 runtime kernel. The hybrid-model Quamba2 comparison is at the accessible block-output surface, not at every internal SSM state tensor. Granite-Small has the densest FFT and KL trace, while DeepSeek and Falcon have narrowed KL/FFT replications; Nemotron dense diagnostics are deferred because throughput would have dominated the Stage 1 budget. Qwen3 scale-up is also deferred because the preregistered W4A16 checkpoint snapshot was not present locally. The M11b positive result is cross-model but not model-invariant: Nemotron top-10 succeeds, while Granite's strongest signal is top-5 with wide uncertainty.

8 Conclusion

Long-reasoning W4A16 channel protection fails when static protected sets drift, and our experiments show that this drift is substantial across models and prompts. Budget-tuned smoothing remedies part of the failure: Nemotron top-10 succeeds, Granite top-5 remains uncertain, and ParoQuant is stronger on Granite. The failed composition result points away from naive stacking and toward architecture-aware method selection as the next positive-method target.

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Ethics Statement

This work analyzes benchmark prompts and model activations. It does not use private user data or make deployment-safety claims. The main ethical risk is overclaiming a scoped mechanism study as a universal quantization recipe; the paper separates measurement, mechanism, and partial-remedy claims.

Reproducibility Statement

The release package contains the public FAST_VERIFY interface, including release/README.md, release/docs/reproducing_results.md, and release/src/scripts/reproduce_all.sh. FAST_VERIFY replays frozen paper-claim values and validates the documented tolerance mapping. Full-fidelity GPU reruns are not implemented inside release/; they remain in the archived experimental packet runners and are disclosed as a release limitation. Stage 1 claim-bearing packets now include the MATH-500 drift replication, DeepSeek/Falcon KL/FFT diagnostic, Granite composition test, and BCa/Holm correction. The format-axis and Qwen3 packets are retained only as infrastructure provenance.

Model	HuggingFace identifier	Snapshot commit
Granite-Small	ibm-granite/granite-4.0-h-small	b8c0982b
Nemotron-3	nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16	cbd3fa9f
DeepSeek-R1-Distill	deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B	ad9f0ae0
Falcon-H1	tiiuae/Falcon-H1-0.5B-Instruct	8f2587ca

Table 2: Model checkpoints used by the experiments. Full provenance and file-level hashes are recorded in the archived result manifests; the release package records claim-to-script tolerance mappings.

Bootstrap seeds, deterministic trace indices, method budgets, and tolerance windows are specified in the release configs. GPU kernels can still exhibit minor nondeterminism across driver versions; the verification compares outputs to bootstrap CI tolerances rather than bit-exact scores. The original Qwen3.6 exploratory hook surface was not used for paper claims because activation access was incomplete.

LLM Use Disclosure

Following the COLM 2026 LLM-use policy adopted from ICLR 2026, we disclose substantive use of large language models in preparing this work. Anthropic Claude and the Codex autonomous coding agent were used to draft and revise paper prose, including the abstract, related work, methods, and results sections; to generate the matplotlib code used for Figures 1–4; to conduct iterative simulated peer review and audit passes; to summarize local experimental artifacts; and to propose, prioritize, and critique candidate experimental methods and follow-up branches, including method families reported in this paper. Codex was also used as a programming assistant for experiment runners, checkers, release-package code, and verification scripts. The human author retained responsibility for final experiment authorization, numerical interpretation, scope decisions, and submission framing. All paper references and attributed prior-work claims were checked against primary sources in the hallucination audit summarized in Appendix D; a later user-flagged scoop check added ResQ, Salfati's KV-cache analysis, and CMPQ after arXiv verification. All reported experimental numbers were traced to committed score caches or result artifacts rather than accepted from LLM output.

Paper description	Internal study label	Provenance location
Granite-Tiny pilot drift measurement	Phase 0	experimental/outlier_migrate/phase0/
Granite-Small drift replication	Phase 1	experimental/outlier_migrate/phase1/
Nemotron-3 drift replication	Phase 2	experimental/outlier_migrate/phase2/
Static migration-aware union, Granite-Tiny	Phase 3	experimental/outlier_migrate/phase3/
Static migration-aware union, Granite-Small	Phase 4	experimental/outlier_migrate/phase4/
DeepSeek Transformer control	Phase 5 prime	experimental/outlier_migrate/phase5_prime/
Falcon-H1 parallel-hybrid control	Phase 7	experimental/outlier_migrate/phase7/
Channel-set methods and baselines	Method study	experimental/outlier_migrate/phase9/
MATH-500 drift replication	Stage 1 E2	experimental/outlier_migrate/phase9/results/om_stage1_e2_mathonly_20260526T1202Z/
Cross-model KL/FFT diagnostic	Stage 1 E1	experimental/outlier_migrate/phase9/results/om_stage1_e1_deepseek_falcon_20260526T2335Z/
ParoQuant+M11b composition	Stage 1 E3	experimental/outlier_migrate/phase9/results/om_stage1_e3_granite_20260526T034442Z/
Format-axis probe	Stage 1 E4	experimental/outlier_migrate/phase9/results/om_stage1_e4_granite_probe_20260527T093400Z/
Qwen3 scale-up probe	Stage 1 E5	experimental/outlier_migrate/phase9/results/om_stage1_e5_qwen3_skipinfra_20260527T093749Z/
BCa/Holm correction	Stage 1 E15	experimental/outlier_migrate/phase9/results/om_stage1_e15_bca_holm_20260527T093931Z/
Paper release interface	Release package	release/docs/reproducing_results.md

Table 3: Translation from paper-facing experiment names to internal study labels. Exact packet directories, command lines, prompt hashes, and model file hashes are maintained in the release documentation and result manifests.

A Detailed Experimental Provenance

The valid Nemotron budget-tuning result uses the corrected static top-1% baseline rerun and reuses validated score caches for the budget arms. The earlier invalid baseline is excluded from all recovery claims. The release mapping table records both the valid rerun and the excluded failed attempt so that reviewers can audit the correction.

Stage 1 packets E1, E2, E3, and E15 are claim-bearing. E4 and E5 are infrastructure-disclosure packets only: E4 did not produce benchmark accuracy/throughput/VRAM rows, and E5 was skipped because the preregistered local Qwen3 W4A16 snapshot was unavailable.

B Per-Method Detailed Results

Experiment	Included traces	No-gap fraction	Median recovery	95% CI
Static union, Granite-Tiny	24	0.417	0.000	[0.000, 0.711]
Static union, Granite-Small	24	0.375	0.000	[0.000, 0.0695]
Position-conditioned channels	8	0.333	-0.867	[-4.25, 0.357]
Position-binned scales	12	0.000	0.234	[-3.47, 0.490]
EMA-smoothed top-1%	8	0.333	0.048	[-8.94, 0.448]
Activation+key coupling	8	0.333	-0.344	[-14.1, 0.731]
DecDEC proxy	8	0.333	-0.070	[-8.50, 0.391]
Budget EMA top-5, Granite	8	0.333	0.449	[-2.28, 1.00]
Static top-10, Nemotron	10	0.167	0.594	[0.246, 0.814]
Budget EMA top-5, Nemotron	10	0.167	0.457	[0.313, 0.760]
Budget EMA top-10, Nemotron	10	0.167	0.815	[0.158, 0.906]
Stable-core protection	8	0.333	0.178	[-1.78, 0.986]
ParoQuant rotation	8	0.333	0.754	[0.477, 1.00]
ParoQuant+M11b top-10	8	0.333	0.565	[-63.6, 0.932]

Table 4: Detailed rounded outcomes for all method families discussed in the main text. Values are recovery ratios relative to each experiment's BF16-vs-static baseline.

Model	AIME set-leaving	MATH-500 set-leaving	Delta
Granite-Small	0.566	0.561	-0.005
DeepSeek-R1-Distill	0.671	0.651	-0.020
Falcon-H1	0.674	0.675	0.002

Table 5: Stage 1 E2 cross-prompt drift replication. MATH-500 rates are computed on 30 prompts per model and all fall within the preregistered ±0.15 tolerance from the AIME-2025 reference.

Model	Layer type	Layers	Strict set-leaving	Within-set shuffling
Granite-Tiny	Attention	4	0.618	0.186
Granite-Tiny	SSM/Mamba	36	0.636	0.174
Granite-Small	Attention	4	0.557	0.282
Granite-Small	SSM/Mamba	36	0.567	0.270
Nemotron-3	Attention	6	0.563	0.255
Nemotron-3	MoE	23	0.527	0.278
Nemotron-3	SSM/Mamba	23	0.533	0.264

Table 6: Layer-type decomposition. Strict set-leaving is the component most directly relevant to protected-set membership; within-set shuffling motivates scale-refresh mechanisms but is not itself membership loss.

C Extended Layer-Stratified Results

D Hallucination and Source Audit

Before finalizing this draft, we audited every bibliography entry against its primary source URL and checked the attributed prior-work claims in the main text. The audit corrected three bibliographic fields: the TechCrunch inference cost article author, the MarketsandMarkets AI-inference report title, and the ParoQuant venue/year label. It also added an explicit citation for test-time TTQ and tightened the wording for Quamba2, ParoQuant, Rotated Runtime Smooth, and SLQ so that the prose matches the source claims. A post-audit scoop addendum added ResQ, Salfati's KV-cache compression analysis, and CMPQ after fetching their arXiv abstract pages; it also corrected the user-supplied ResQ attribution from "Saha et al." to the verified Saxena et al. author list. No unverified bibliography entries remain after these corrections. We separately re-derived the headline experimental numbers from local score caches and result manifests; the audit identified only wording/rounding fixes, not a change to any reported result. The full audit table is maintained in docs/hallucination_audit_report.md.